

Split Complex Lorentzian Dynamics

Part XXIII: Local Sections, Transition Signs, and Exact Reconstruction of the Quaternionic Double Cover

From a certified global projection to intrinsic local gluing, route-independent reconstruction, and a basis-free frame-level target

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Abstract

Part XXI ended with a certified group-theoretic picture: an intrinsic internal carrier $\mathcal{L}$ equivalent to unit quaternions and hence classically to $SU(2) \cong Spin(3)$, a visible group G_{vis} equivalent to $SO(3)$, a certified projection $p : \mathcal{L} \rightarrow G_{\text{vis}}$ identified pointwise with quaternionic rotation and having kernel $\{\pm 1\}$, and a larger continuously central carrier $\mathcal{L}_Z$ whose normalized factor is abstractly equivalent to $U(2)$ [6, 1]. Those results answered the recognition problem but left a categorical gap: the project knew the total group, the visible group, and the covering homomorphism, yet had not reconstructed the local data by which such a global object is assembled.

Task 23 returns deliberately to decomposition and reconstruction. It freezes the certified projection and, without using covering-space vocabulary in the reconstruction layer, constructs four intrinsic open subsets $V_i \subset G_{\text{vis}}$ from inherited coordinates of the internal carrier. The four sets cover the visible group and form a minimal subfamily of that coordinate construction. On each V_i , the source constructs a continuous local representative s_i as the inverse of an explicit continuous open bijection, normalized by positivity of one inherited coordinate. On overlaps, two local representatives differ by a unique kernel sign

$$s_j(g) = \tau_{ij}(g) s_i(g), \quad \tau_{ij}(g) \in \{\pm 1\}.$$

The functions τ_{ij} are continuous and therefore locally constant, and satisfy the exact identity, inverse, and triple-overlap laws.

A central tension appears at this point. The tempting simplification to one fixed sign for each pair (i, j) is false. Task 23 preserves an explicit counterexample inside $V_0 \cap V_1$: two visible points give opposite transition signs. Consequently that overlap is not preconnected, and the correct transition datum is a locally constant sign function, equivalently a sign on each connected component, not one sign per pair of charts. This failed intended theorem is not discarded; it becomes the theorem that fixes the correct local geometry.

With the corrected transition data, the development proves local product homeomorphisms

$$p^{-1}(V_i) \cong V_i \times \{\pm 1\},$$

then forms the disjoint union of the four local products and glues them by the transition signs. The quotient $\mathcal{L}_{\text{loc}}$ is proved homeomorphic to the original certified carrier $\mathcal{L}$, and the homeomorphism recovers the projection as well. Thus the local sections and their $\{\pm 1\}$ -valued overlap functions are not merely compatible with the global carrier: they are exactly sufficient to reconstruct it.

The old reconstruction layers are then revisited. The earlier direction domains $\text{Dom}(v)$ are not the new visible-group charts: an extra angle coordinate and direction reversal are needed, and the new domain V_0 has no old $\text{Dom}(v)$ counterpart. Old core Bridges do reduce exactly to the new overlap signs, while the extended carrier retains central factors outside $\{\pm 1\}$ and

has an infinite fibre. Integer information also bifurcates: shifting the angle parameter by k full turns acts on the core by $(-1)^k$, whereas an integer central-label difference evaluates to $+1$ after every full turn. Hence parity is the correct reduction for the parameter-shift action but not for the central-label full-turn map.

The hierarchy is now exact. The visible group alone is insufficient; visible group plus local sections with trivial transitions is still insufficient; visible group plus local sections plus the locally constant $\{\pm 1\}$ -transition functions is sufficient to reconstruct the core carrier. The same data are provably insufficient for $\mathcal{L}_Z$. Finally, only after all local ingredients have been reconstructed does the comparison module invoke Mathlib’s standard definition and prove `IsCoveringMap pr`. The object is therefore formally identified as a two-sheeted covering map, but no frame bundle, spin structure, characteristic class, or physical spin degree of freedom is constructed.

Task 24 should take one carefully bounded step upward, but it must preserve the same dependency discipline. It should not define the next object from the already identified group or from the four Task-23 charts. Instead it should begin with the certified three-dimensional visible carrier, its positive-definite inner product, and its inherited orientation, reconstruct the space of oriented orthonormal frames intrinsically and without a chosen basis, and only afterwards prove that the visible group acts freely and transitively on that frame space. A reference frame may enter only as a later coordinate choice whose change-of-reference equivalence is proved. The certified double cover can then be tested for a reference-independent two-sheeted lift of this frame torsor. This is the smallest frame-level object on which a genuine lift question can be formulated without assuming a manifold-level spin structure in advance.

Keywords: split-complex Lorentz structure; unit quaternions; $SU(2)$; $SO(3)$; $Spin(3)$; double cover; local sections; transition signs; gluing; covering map; frame torsor; Lean 4; formal verification; Conservation of Difficulty.

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1 From recognition back to reconstruction

The sequence from Part XIX through Part XXI was an explanatory detour forced by success. Earlier tasks had reconstructed a large collection of local exact representatives, central Bridges, half-odd labels, reversal constraints, and global obstruction statements. Part XIX showed that these objects had become too tightly coupled to continue safely without knowing what they collectively were. Part XX located a unit-quaternion core. Part XXI, completed and certified by Task 22, then fixed the main group-level identifications: the internal core is equivalent to $SU(2)$, the visible group is equivalent to $SO(3)$, and the intrinsic projection is the quaternionic rotation map with kernel $\{\pm 1\}$ [6, 1].

That resolution creates a new methodological danger. Once a classical object has been recognized, it becomes easy to import its standard theory and thereby lose the dependency structure that the project is attempting to expose. Task 23 therefore refuses the obvious shortcut. It does not begin by saying “take the standard double cover and choose local trivializations.” Instead it freezes only the already certified map and reconstructs the local structure from the inherited coordinates of the project.

Methodological principle 1 (Return to decomposition). A classical identification may be used as a certified boundary condition, but it does not replace reconstruction. The next layer must show which local data are forced by the existing carrier and projection, and whether those data are sufficient to recover the global object.

A second distinction is equally important. The structure of the object must be separated from a particular route used to expose it. The four domains constructed below are an explicit successful local witness family for the certified projection; they are not promoted into the definition of the projection or of a future frame object. Likewise, a later choice of basis or reference frame may coordinatize an intrinsic object but must not define that object by fiat.

Methodological principle 2 (Object structure versus reconstruction route). Keep two dependency ledgers separate. The first records what belongs to, is required by, or is reconstructed from the intrinsic object. The second records auxiliary witnesses, charts, coordinates, normalizations, or lemmas used by one proof route. An arrow in the second ledger must not be silently reversed into a constituent of the first.

This distinction will matter especially after Task 23. The coordinate cover proves local triviality and supports exact reconstruction, but the resulting covering map is the invariant object. Future tasks should therefore reconstruct the next carrier basis-free first and only then compare it with convenient group or coordinate models.

The narrow target is therefore not a new group. It is the already known map

$$p : \mathcal{L} \longrightarrow G_{\text{vis}}, \tag{1}$$

viewed as an object to be decomposed.

2 Formal provenance and verification

Task 23 is a complete green run. The supplied archive corresponds to Aristotle run

c3139bdc-c8f5-4e55-83bc-eeb637a826dc

and has SHA-256

c8312fe9dcdf5d76b623066ffd73ba944a96a751f87aa3dc1fa363bfb5ee6097.

The formal environment is Lean 4.28.0 with Mathlib resolved at revision

8f9d9cff6bd728b17a24e163c9402775d9e6a365.

The Task-23 layer contains twelve Lean modules in a strict linear import chain:

SafeBase → Cover → Sections → Transitions → LocalProduct → Reconstruction
 → GlobalSection → OldDomains → OldBridges → Hierarchy → Comparison → Task23. (2)

The source totals 2592 lines. `Task23.lean` introduces no new mathematics; its fifty-five principal endpoints are aliases of the declarations proved in the preceding modules. The Task-21/22 source is imported once, as fixed certified input, and was not modified [2].

Check	Result	Detail
Focused Task-23 build	PASS	<code>lakebuildRequestProject.NullSectorTask23.Task23</code> , 8205 jobs
Whole-project build	PASS	<code>lake build</code> , 8276 jobs
Task-23 build errors	0	both final builds green
Task-23 warnings	0	114 project warnings all pre-existing
Principal endpoints	55	all exposed by aliases
Axiom set	exact baseline	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>sorry / admit</code>	0 / 0	no proof holes
custom axioms / proof escapes	0	no <code>implemented_by</code> , <code>native_decide</code> , <code>bv_decide</code> , <code>unsafe</code> , or <code>partial</code>

The build emits 139 axiom-report lines because earlier imported endpoint audits are also present, but every line reports exactly the same three-axiom baseline. In particular, `sorryAx` is absent.

3 Package A: freeze the projection and expose only the kernel sign

Task 23 begins by forgetting the newly learned conventional names. It retains only the certified carrier, visible carrier, projection, and the two-element kernel action. In Lean notation the basic carriers are

$$\text{LiftT} := \mathcal{L}, \quad (3)$$

$$\text{GvisT} := G_{\text{vis}}, \quad (4)$$

$$\text{Sgn} := \{r \in \mathbb{R} : r = 1 \text{ or } r = -1\}. \quad (5)$$

The sign carrier inherits multiplication and is proved to be a finite commutative group with discrete topology. Its action on $\mathcal{L}$ is scalar multiplication, written here as

$$\varepsilon \cdot u, \quad \varepsilon \in \{\pm 1\}, \quad u \in \mathcal{L}.$$

The first exact fibre theorem is

Theorem 1 (Certified fibre classification). *For all $u, v \in \mathcal{L}$,*

$$p(u) = p(v) \iff \exists \varepsilon \in \{\pm 1\} : v = \varepsilon \cdot u, \quad (6)$$

and the sign ε is unique.

This theorem is the local algebraic engine for the rest of the task. It says that every visible fibre contains exactly the two kernel-related representatives and nothing else. At this stage no covering-space terminology is used.

4 Package B: four intrinsic local domains

The local cover is not imported from a standard atlas of $SO(3)$. It is built from four coordinates already present in the ambient carrier W :

$$q_i(u) = (u_0, -u_6, u_5, -u_4)_i, \quad i \in \{0, 1, 2, 3\}. \quad (7)$$

For an internal element define

$$U_i = \{u \in \mathcal{L} : q_i(u) \neq 0\},$$

and push it to the visible carrier,

$$V_i = p(U_i) \subset G_{\text{vis}}. \quad (8)$$

Because $q_i(-u) = -q_i(u)$, nonvanishing is invariant under the kernel sign, so V_i is well-defined on visible elements.

The key quadratic identity on $\mathcal{L}$ is the four-coordinate sphere relation, from which one deduces that every internal element has at least one nonzero q_i . Therefore

$$\bigcup_{i=0}^3 V_i = G_{\text{vis}}. \quad (9)$$

Each V_i is open. Moreover, the source constructs a basis element for every coordinate and proves that its visible image lies in V_i and in no other V_j . Hence no proper subfamily of these four coordinate domains covers G_{vis} .

Remark 1 (Meaning of minimality). *The four-domain result is a minimality statement for this intrinsic coordinate family. It is not a theorem that no different three-chart cover of G_{vis} exists.*

This distinction will matter again when disconnected overlaps appear.

5 Package C: local sections from positivity, not choice

The nonzero coordinate condition still leaves the two kernel-related representatives. Task 23 removes that ambiguity locally by choosing the representative with positive coordinate:

$$U_i^+ = \{u \in \mathcal{L} : 0 < q_i(u)\}. \quad (10)$$

The restricted projection

$$p_i^+ : U_i^+ \longrightarrow V_i$$

is proved continuous, open, injective, and surjective. It therefore yields a homeomorphism

$$U_i^+ \cong V_i. \quad (11)$$

The local section is defined as its inverse:

$$s_i : V_i \longrightarrow \mathcal{L}, \quad s_i = (p_i^+)^{-1}. \quad (12)$$

Consequently

$$p(s_i(g)) = g, \quad (13)$$

$$q_i(s_i(g)) > 0, \quad (14)$$

and s_i is continuous by construction.

This point is methodologically important. A global set-theoretic representative had already been known to exist. Task 23 does not restrict such a global choice and attempt to prove it continuous. It constructs the local representative directly from the intrinsic coordinate normalization.

The opposite local normalization is simply

$$s_i^-(g) = -s_i(g).$$

More generally, any local representative over V_i differs pointwise from s_i by a unique sign.

6 Packages D and E: the transition signs

On a nonempty overlap $V_i \cap V_j$, the two local sections have the same visible image. By the fibre theorem there exists a unique $\{\pm 1\}$ -valued relative factor. Task 23 defines

$$\tau_{ij}(g) \in \{\pm 1\} \tag{15}$$

by the exact equation

$$s_j(g) = \tau_{ij}(g) s_i(g). \tag{16}$$

The source derives τ_{ij} from the sign of the inherited coordinate $q_j(s_i(g))$. Thus the transition function is not inserted as a conventional bundle datum; it is reconstructed from the already existing local representatives.

The transition functions are continuous. Since $\{\pm 1\}$ has discrete topology, they are locally constant. Task 23 then proves, in the general order expected of relative factors,

$$\tau_{ii}(g) = +1, \tag{17}$$

$$\tau_{ji}(g) = \tau_{ij}(g)^{-1}, \tag{18}$$

$$\tau_{ik}(g) = \tau_{jk}(g) \tau_{ij}(g). \tag{19}$$

Only after equation (18) is established does the order-two property simplify it to $\tau_{ji} = \tau_{ij}$.

At this stage the equations have exactly the standard shape of transition-function or cocycle identities. Task 23 deliberately does not introduce a cohomology class.

7 The failed simplification that reveals the correct geometry

A natural next move would be to assign one sign to each unordered pair of domains. That would be valid if every overlap were preconnected. Task 23 attempts precisely this simplification and discovers that it is false.

The source constructs two explicit internal elements with quaternion-coordinate pattern

$$\left(\frac{\sqrt{2}}{2}, \pm \frac{\sqrt{2}}{2}, 0, 0 \right) \tag{20}$$

in the relevant four-coordinate representation. Their visible images both lie in $V_0 \cap V_1$, but

$$\tau_{01}(g_+) = +1, \quad \tau_{01}(g_-) = -1.$$

If the entire overlap were preconnected, local constancy would force these values to agree. Therefore

$$V_0 \cap V_1 \text{ is not preconnected.} \tag{21}$$

Theorem 2 (Correct transition-data boundary). *The Task-23 cover does not admit one transition sign per pair of domains. The correct datum is a locally constant function*

$$\tau_{ij} : V_i \cap V_j \rightarrow \{\pm 1\},$$

or equivalently a constant sign on each preconnected component of the overlap.

This is the main negative result of the task, and it improves rather than weakens the reconstruction. The failed theorem identifies exactly which local information must be retained for the global gluing to work.

8 Package F: local product homeomorphisms

The local section and sign action now determine an explicit product map

$$V_i \times \{\pm 1\} \longrightarrow p^{-1}(V_i), \quad (g, \varepsilon) \longmapsto \varepsilon s_i(g). \quad (22)$$

Its inverse sends $u \in p^{-1}(V_i)$ to

$$(p(u), \text{sgn}(q_i(u))).$$

Task 23 proves both inverse identities, continuity of both maps, and compatibility with the base projection. The result is the homeomorphism

$$p^{-1}(V_i) \cong V_i \times \{\pm 1\}. \quad (23)$$

This is the first point in the cumulative project where the double structure has been reconstructed as an actual local product rather than recognized from kernel cardinality or periodicity.

9 Package G: reconstruction from local pieces

The principal endpoint of Task 23 asks whether the local data are sufficient, not merely necessary.

Form the disjoint union

$$\mathcal{P} = \coprod_{i=0}^3 (V_i \times \{\pm 1\}). \quad (24)$$

A point of $\mathcal{P}$ is written (i, g, ε) . The local section assembles it into an internal element

$$\text{glue}(i, g, \varepsilon) = \varepsilon s_i(g). \quad (25)$$

Two local data are declared equivalent when they describe the same base point and their signs are related by the appropriate transition function:

$$(i, g, \varepsilon) \sim (j, g, \eta) \iff \eta = \tau_{ij}(g)\varepsilon. \quad (26)$$

Reflexivity follows from $\tau_{ii} = 1$, symmetry from the inverse law, and transitivity from the triple-overlap law. Hence this is an equivalence relation.

Define the reconstructed carrier

$$\mathcal{L}_{\text{loc}} = \mathcal{P} / \sim. \quad (27)$$

Because equation (26) is exactly equality after the gluing map, glue descends to a natural map

$$\Phi : \mathcal{L}_{\text{loc}} \longrightarrow \mathcal{L}.$$

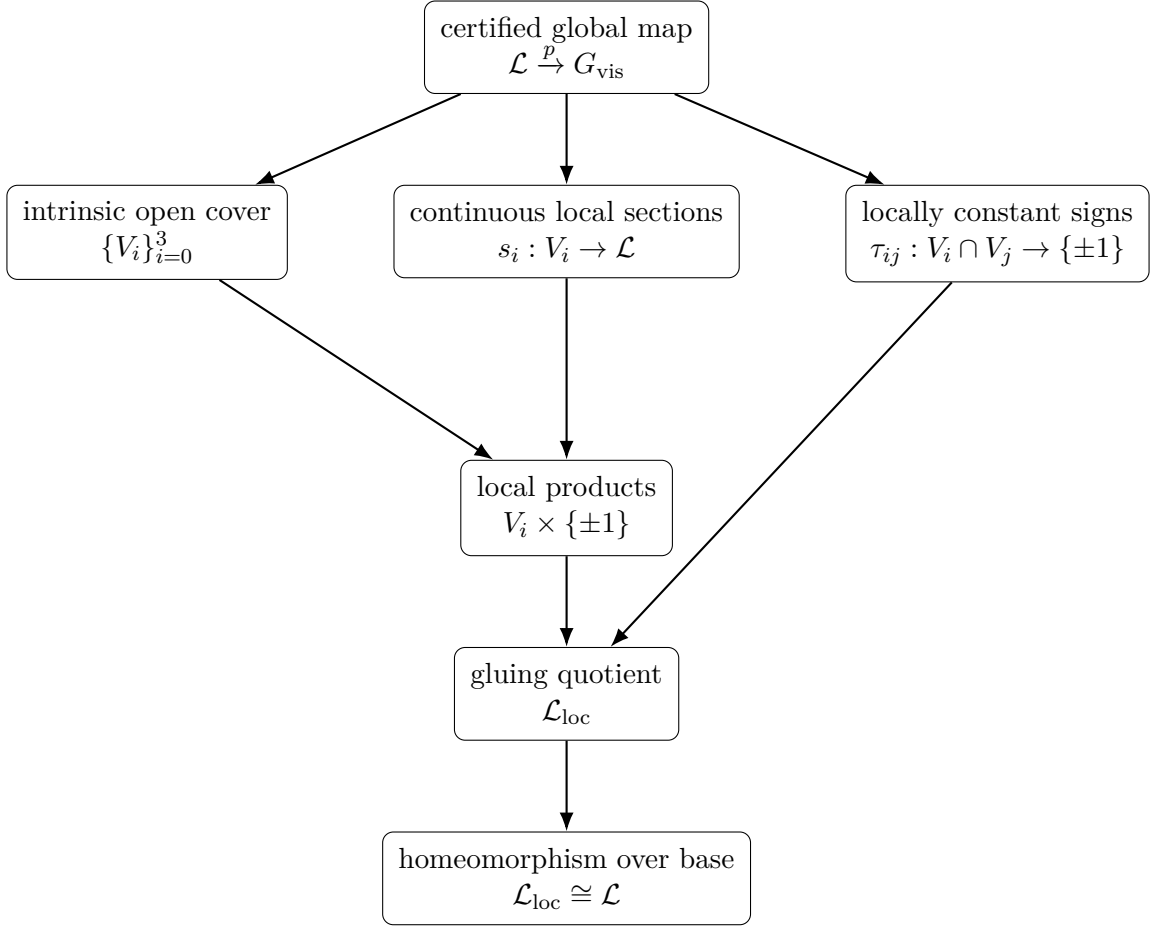
Task 23 proves Φ injective and surjective. It then proves continuity and openness, obtaining the strongest topological result:

Theorem 3 (Exact local-to-global reconstruction). *There is a homeomorphism*

$$\mathcal{L}_{\text{loc}} \cong \mathcal{L} \quad (28)$$

under which the reconstructed base map agrees exactly with the certified projection p .

Thus the transition signs are not decorative. They contain precisely the missing global information that turns four trivial local products into the original nontrivial total carrier.



10 Package H: what became of the old $\text{Dom}(v)$ domains?

The earlier Tasks 16–19 had already produced domains $\text{Dom}(v)$, but these live in unit-direction space, not in the visible transformation group. Task 23 therefore asks for an exact comparison instead of identifying them by analogy.

For a unit direction n and parameter θ , let $\text{vis}(n, \theta) \in G_{\text{vis}}$ denote the visible transformation generated by the old reference implementer. In the inherited quaternion coordinates one has

$$q_0(U_n(\theta)) = \cos(\theta/2), \quad (29)$$

$$q_1(U_n(\theta)) = \sin(\theta/2)n_1, \quad (30)$$

$$q_2(U_n(\theta)) = \sin(\theta/2)n_2, \quad (31)$$

$$q_3(U_n(\theta)) = \sin(\theta/2)n_3. \quad (32)$$

Consequently

$$\text{vis}(n, \theta) \in V_0 \iff \cos(\theta/2) \neq 0, \quad (33)$$

while, for the first spatial coordinate,

$$\text{vis}(n, \theta) \in V_1 \iff \sin(\theta/2)n_1 \neq 0. \quad (34)$$

Task 23 rewrites equation (34) exactly as

$$\text{vis}(n, \theta) \in V_1 \iff \left(n \in \text{Dom}(e_1) \text{ or } -n \in \text{Dom}(e_1) \right) \wedge \sin(\theta/2) \neq 0. \quad (35)$$

So the old domain appears, but only after two additional ingredients are restored: the parameter and the reversal ambiguity.

Where the sign in equation (30) is positive, the old reference implementer is exactly the new normalized section. Hence the old locally exact data genuinely induce part of the new atlas.

But the correspondence is not exhaustive. The domain V_0 depends only on $\cos(\theta/2)$, and Task 23 proves that there is no vector v for which membership in V_0 is equivalent to $n \in \text{Dom}(v)$ for all axis-angle presentations. The proof uses the identity element, which lies in V_0 for both n and $-n$, whereas no old $\text{Dom}(v)$ can contain an antipodal pair.

Status 1 (Old-domain verdict). The old $\text{Dom}(v)$ were genuine precursors of some local representative domains, but they were not already the atlas of the certified double cover. The visible-group atlas requires parameter information and kernel/reversal identifications that the direction-only domains did not contain.

11 Packages I and J: Bridges, signs, and the two meanings of an integer

Task 23 next revisits the old relative Bridge data. The result separates the core and extended carriers more sharply than before.

11.1 Core Bridge: exact reduction to a sign

If $u, v \in \mathcal{L}$ satisfy $p(u) = p(v)$, then there exists exactly one $\varepsilon \in \{\pm 1\}$ with

$$v = \varepsilon u. \quad (36)$$

Thus, after restricting to the certified core, the old relative Bridge has no residual continuous freedom: it is exactly the transition sign.

11.2 Extended Bridge: the sign is insufficient

For the larger carrier $\mathcal{L}_Z$, Task 23 constructs an explicit central element corresponding to the inherited S -direction which is invertible, invisible under the visible projection, and unequal to both $+1$ and -1 . Multiplying a core element by this factor leaves the visible transformation unchanged but takes the result outside the core carrier.

The extended kernel also contains an injective continuous one-parameter family, so its fibre over the identity is infinite. Therefore the $\{\pm 1\}$ -transition data that reconstruct $\mathcal{L}$ cannot reconstruct $\mathcal{L}_Z$.

11.3 Parameter-shift parity versus central-label full turns

The most delicate correction concerns integer labels. Define the core parity map

$$\pi(k) = (-1)^k \in \{\pm 1\}, \quad k \in \mathbb{Z}. \quad (37)$$

Task 23 proves that shifting the axis-angle parameter by k full turns changes the internal reference representative exactly by $\pi(k)$. Here parity is the correct integer-to-sign reduction.

The old central label difference behaves differently. For the integer central Bridge, evaluation after one full turn gives

$$\text{labelBridge}(m, 2\pi) = +1 \quad \text{for every } m \in \mathbb{Z}. \quad (38)$$

This map is constant, not parity. Distinct integer labels can therefore agree both in core projection and at full-turn sign while differing at other parameter values.

Theorem 4 (Two integer reductions). *There are two distinct integer-to-kernel mechanisms in the cumulative construction:*

1. *full-turn shifts of the angle parameter reduce by parity $k \mapsto (-1)^k$;*
2. *full-turn evaluation of an integer central label difference is identically $+1$.*

They must not be identified.

This distinction resolves an ambiguity that remained natural before the core and extended structures had been separated completely.

12 Package K: the minimal reconstruction hierarchy

Task 23 asks not only whether one reconstruction works, but how much information is necessary.

Level	Data	Reconstructs the core?
1	visible group G_{vis} alone	NO
2	visible group plus local sections, with transition data discarded/trivialized	NO
3	visible group plus local sections plus the locally constant $\{\pm 1\}$ -transition functions	YES
4	old integer labels and continuous central Bridges in addition	more information than needed for $\mathcal{L}$, still a separate requirement for $\mathcal{L}_Z$

Levels 1 and 2 are refuted using a trivial comparison total space $G_{\text{vis}} \times \{\pm 1\}$. It has the same two-element fibres and admits continuous local sections with trivial transitions, but it also has a global continuous section, whereas the certified carrier does not. Thus the transition data cannot be inferred from the base plus the existence of local sections.

Level 3 is exactly theorem 3. It is sufficient.

For the extended carrier the fibre over the identity is infinite, while the core fibre has exactly two elements. Hence any local reconstruction of $\mathcal{L}_Z$ must retain additional continuous central data.

Methodological principle 3 (Location of the global information). For the core carrier, the nontrivial global information is neither in the visible base nor in the existence of local representatives. It is in the way those representatives are glued.

13 Package L: the global no-go read entirely from local data

Earlier tasks had proved the nonexistence of a global continuous representative. Task 23 now derives the exact local equivalent.

Let $c_i : V_i \rightarrow \{\pm 1\}$ be continuous local sign modifications and define

$$s_i^c(g) = c_i(g)s_i(g).$$

The family trivializes the transitions if the modified sections agree on every overlap. Task 23 proves the equivalence

Theorem 5 (Global section versus transition trivialization). *There exists a global continuous section $\sigma : G_{\text{vis}} \rightarrow \mathcal{L}$ of p if and only if there exists a continuous local sign family $\{c_i\}$ that makes all modified overlap transitions trivial.*

Both implications are proved directly from the reconstructed local data. In one direction, compare the global section with each s_i and extract its local sign. In the other, patch the modified sections over the open cover.

Because the certified global section does not exist, the theorem yields an unavoidable local obstruction:

$$\text{every continuous local sign modification leaves at least one nontrivial overlap sign.} \quad (39)$$

Yet a set-theoretic trivializing family does exist. The obstruction is therefore topological, not merely combinatorial or algebraic.

This local reformulation is one of the most useful consequences of Task 23: the global no-go can now be read directly from the transition data rather than from an externally recognized double-cover theorem.

14 Comparison only: the covering-map theorem

Only after Packages A–L have produced all of the required local ingredients does the final comparison module import the conventional vocabulary. It collects eight facts:

1. p is a continuous surjection;
2. every fibre is a discrete two-element sign fibre;
3. the V_i form an explicit open cover;
4. every V_i carries a continuous local section;
5. every preimage $p^{-1}(V_i)$ is homeomorphic over the base to $V_i \times \{\pm 1\}$;
6. transition functions are continuous and locally constant;
7. identity, inverse, and triple-overlap laws hold;
8. the global carrier is recovered by gluing the local pieces.

The module then proves, using Mathlib’s own standard definition,

$$\text{IsCoveringMap } p. \quad (40)$$

Thus the project does not infer a cover from a two-element kernel. It reconstructs the complete local structure and only then recognizes the standard object [7, 3].

Proof boundary 1 (Exact conventional conclusion). *Task 23 proves that $p : \mathcal{L} \rightarrow G_{\text{vis}}$ is a two-sheeted covering map. It does not construct a frame bundle, tangent bundle, Lorentz-frame bundle, principal spin bundle, spin structure, characteristic class, or cohomology class.*

15 How much closer is this to spin geometry?

The answer is now more precise than in Part XXI.

At the group level, the project has the classical substrate

$$\mathcal{L} \simeq SU(2) \cong Spin(3), \quad G_{\text{vis}} \simeq SO(3),$$

with the quaternionic double-cover map and kernel $\{\pm 1\}$. Task 23 adds the entire local topological structure of that cover: local sections, local product homeomorphisms, transition functions, and gluing reconstruction.

This is exactly the infrastructure that a spin lifting problem uses at the structure-group level. But a spin structure is not a property of the group homomorphism alone. In the standard geometric definition one begins with an oriented orthonormal frame bundle over a base and asks for a compatible lift through the spin-group covering map [4, 5]. No such base-dependent frame bundle exists in the present formal layer.

Status 2 (Spin boundary after Task XXIII). • spin-group carrier: effectively reached at the certified group level;

- standard double-cover group homomorphism: certified;
- local two-sheeted covering structure and transition signs: certified;
- a frame space over a nontrivial base: not reconstructed;
- a lift of a frame bundle: not reconstructed;
- a spin structure: not reconstructed.

The Conservation of Difficulty has therefore moved again. The problem is no longer “what is the cover?” or “how is the cover glued?” Those questions are closed. The missing object is now the geometric carrier to which the cover would have to be applied as a lift.

16 Conservation of Difficulty after Task XXIII

The cumulative migration of difficulty is now unusually clean:

$$\begin{array}{l}
 \text{local exact representatives and half-angle structure} \\
 \Downarrow \\
 \text{global incompatibility and relation-completeness problem} \\
 \Downarrow \\
 \text{unit-quaternion core and } SU(2) \rightarrow SO(3) \text{ recognition} \\
 \Downarrow \\
 \text{certified double cover with kernel } \{\pm 1\} \\
 \Downarrow \\
 \text{intrinsic local sections and locally constant transition signs} \\
 \Downarrow \\
 \text{exact reconstruction of the global carrier from gluing} \\
 \Downarrow \\
 \boxed{\text{what intrinsic frame-level object is this cover supposed to lift?}}
 \end{array} \tag{41}$$

Four lessons from Task 23 are particularly relevant for the next move.

First, simplification must continue to be tested rather than assumed. The failed connected-overlap expectation was not a nuisance; it exposed the correct local invariant. Any future frame-level reconstruction must therefore preserve componentwise transition data until connectedness is proved.

Second, the local witness family must not be mistaken for the object itself. The four V_i form a minimal cover only within the explicit coordinate family constructed in Task 23. Their success proves the local structure of p ; it does not make four charts part of the definition of the covering map. Another valid local decomposition could expose the same invariant object through a different route.

Third, representation freedom should be removed before new structure is named. In the next step the correct order is therefore not $G_{\text{vis}} \rightarrow \text{frames}$ by definition, but rather

$$E \longrightarrow \text{inner product and orientation} \longrightarrow \mathcal{F}^+(E) \longrightarrow \text{derived } G_{\text{vis}}\text{-action}.$$

Only after $\mathcal{F}^+(E)$ has been constructed basis-free may a reference frame identify it with a group torsor for computational convenience.

Fourth, the project now knows exactly which information is lost by each quotient and which extra datum restores it. The visible projection forgets the kernel sign, and local $\{\pm 1\}$ -transitions restore precisely enough information to reconstruct $\mathcal{L}$. The full-turn sign forgets still more: it does not retain the complete integer/central-Bridge data needed for $\mathcal{L}_Z$. Future quotients should therefore carry an explicit information-loss ledger rather than relying on suggestive notation alone.

A final engineering distinction should also be retained. If a future step cannot be stated or packaged because the pinned library lacks a suitable topological, bundle, or quotient object, that is an infrastructure boundary, not automatically a mathematical obstruction. In that case the smallest explicit intrinsic predicate should be constructed first and compared with standard library structures only afterwards.

17 Task XXIV: basis-free frame-torsor reconstruction

Task 24 should be a new decomposition/reconstruction task, but it should remain narrow. The project is not yet ready for the broad question “does spacetime carry a spin structure?” It first needs to reconstruct, independently of any chosen coordinates, the object that can legitimately play the role of an oriented frame fibre.

The certified input is deliberately smaller than the conventional answer. Let E be the already reconstructed real three-dimensional spatial carrier with its inherited positive-definite inner product and orientation datum. The visible group G_{vis} and the cover $\mathcal{L} \rightarrow G_{\text{vis}}$ are available as theorems, but neither should be used to define the frame carrier itself.

A suitable neutral title is:

Task XXIV – Basis-Free Oriented Frame Reconstruction: The Intrinsic Frame Torsor of the Visible Three-Space and Its Lift through the Certified Double Cover.

The task should be restricted to the following finite chain.

17.1 WP1: reconstruct the oriented frame carrier without a reference frame

Define a frame candidate directly as an ordered triple $(e_1, e_2, e_3) \in E^3$ satisfying the inherited orthonormality relations. Define positive orientation from the determinant/orientation datum already available on E , not by comparison with a permanently distinguished frame. Construct

$$\mathcal{F}^+(E)$$

as the intrinsic carrier of positively oriented orthonormal frames.

Required endpoints:

1. well-defined orthonormality and orientation predicates;
2. nonemptiness of $\mathcal{F}^+(E)$;
3. independence of the carrier definition from any basis or reference-frame choice.

A temporary basis may be used inside a proof only if the final theorem eliminates it.

17.2 WP2: derive the visible-group action rather than build frames from the group

Use the already certified action of G_{vis} on E to obtain a componentwise action on $\mathcal{F}^+(E)$. Prove preservation of orthonormality and orientation.

Then prove or refute:

1. freeness of the action;
2. transitivity of the action.

If both hold, record that $\mathcal{F}^+(E)$ is a G_{vis} -torsor. This conclusion must be derived from the intrinsic frame carrier. Defining $\mathcal{F}^+(E) := G_{\text{vis}}$ is prohibited.

17.3 WP3: recover the group as the difference object of frames

For two frames $f, f' \in \mathcal{F}^+(E)$, prove existence and uniqueness of the visible transformation taking f to f' . This establishes the exact equivalence

$$\text{frame difference} \iff \text{unique element of } G_{\text{vis}}$$

without choosing a global origin in the torsor.

Only after this theorem may a reference frame f_0 be chosen to produce a coordinate equivalence $G_{\text{vis}} \simeq \mathcal{F}^+(E)$. Prove explicitly that replacing f_0 by another reference frame changes only the coordinate identification and not the intrinsic frame object.

17.4 WP4: audit what the frame quotient forgets

Before constructing a lifted frame object, determine exactly what information is lost when the $\mathcal{L}$ -action is projected through $p : \mathcal{L} \rightarrow G_{\text{vis}}$. In particular, prove the role of the kernel sign on frame data and isolate the minimal extra datum required to distinguish the two lifted representatives over one ordinary frame.

This package should continue the information-loss discipline established in Task 23: quotient first, forgotten datum second, reconstruction third.

17.5 WP5: construct the lifted-frame torsor

Using the basis-free frame carrier and the certified double cover, construct the smallest lifted-frame candidate compatible with the derived G_{vis} -torsor structure. A pullback or quotient construction is acceptable only after its defining maps have been reconstructed intrinsically.

Prove or refute:

1. each oriented frame has exactly two lifted representatives;
2. $\mathcal{L}$ acts freely and transitively on the lifted-frame carrier;
3. the projection from lifted frames to ordinary frames is a two-sheeted covering map;
4. any construction that temporarily uses a reference frame is canonically equivalent to the corresponding construction using any other reference frame.

17.6 WP6: compare the Task-23 atlas only after the frame object exists

The four Task-23 domains V_i , their local sections, and their $\{\pm 1\}$ -valued transition functions should enter only at this stage. Pull them to the reconstructed frame torsor and determine whether they provide one valid local decomposition of the lifted-frame projection.

Do not require the frame object to inherit exactly four charts, and do not define its local structure from this particular atlas. The comparison should answer instead whether the previously reconstructed transitions represent the same invariant lift data in frame coordinates.

17.7 WP7: distinguish mathematical and infrastructure boundaries

If the required frame/torsor/lift statements are mathematically meaningful but unavailable as convenient Mathlib package structures, record this separately from theorem failure. Prefer a minimal explicit intrinsic definition and prove its laws before introducing a comparison-only conventional wrapper.

The final audit must classify every unresolved endpoint as one of:

- mathematical obstruction or counterexample;
- missing hypothesis or dependency;
- formalization/library infrastructure boundary.

17.8 Stop condition

Task 24 must stop at a single basis-free frame fibre/torsor and its possible two-sheeted lift. It must not introduce:

- a spacetime manifold or another nontrivial base as new input;
- a tangent or Lorentz-frame bundle over such a base;
- characteristic classes or w_2 ;
- a global spin structure on a manifold;
- dynamics or physical spin observables.

The success criterion is deliberately modest:

Can the already reconstructed metric-oriented three-space generate its own basis-free oriented-frame torsor, can the visible group be recovered as the torsor's difference group rather than assumed as its definition, and does the certified double cover then produce a reference-independent two-sheeted lift of that torsor?

If the answer is yes, Task 24 will supply the first genuinely basis-independent frame-level object of the project. A later task can then perform a separate dependency audit of what additional ingredients would be required to organize such frame fibres over a nontrivial base. That later audit, rather than Task 24, is the appropriate place to ask which dependencies of a genuine frame bundle or spin structure are present, absent, independent, or only library-supported.

18 Conclusion

Task 23 closes the local-structure problem of the certified double cover. It begins with no standard covering-space atlas and no assumed transition cocycle. Four open domains are extracted from inherited internal coordinates. Continuous local representatives are selected by positivity. Their unique relative factors land in the certified kernel $\{\pm 1\}$, are continuous and locally constant, and obey the exact transition laws.

The expected shortcut then fails: one cannot compress every overlap to a single sign. The explicit counterexample in $V_0 \cap V_1$ proves that the overlap is not preconnected. The transition information must remain componentwise. Preserving that extra information is exactly what allows the next step to succeed.

The local products $p^{-1}(V_i) \cong V_i \times \{\pm 1\}$ are proved as homeomorphisms. Their disjoint union, glued by the reconstructed transition signs, produces a quotient carrier $\mathcal{L}_{\text{loc}}$ homeomorphic to the original $\mathcal{L}$, with the projection recovered at the same time. The global object has therefore been decomposed and reconstructed within the formal system.

The comparison with the earlier layers is equally clarifying. The old $\text{Dom}(v)$ domains were not secretly the final atlas; they lacked angle information and reversal identification, and one new domain has no old counterpart. The old core Bridge does become the new transition sign, but the extended Bridge retains continuous central information. The integer story also splits: parameter shifts reduce by parity, central integer-label full turns do not.

Only after all of this is proved does the comparison layer invoke Mathlib and classify p as a covering map. The phrase “double cover” is therefore no longer heuristic, no longer inferred from $2\pi/4\pi$ behavior, and no longer merely a consequence of recognizing $SU(2) \rightarrow SO(3)$. It has been reconstructed from its local pieces and certified as such.

This brings the project materially closer to spin geometry while sharpening the remaining boundary. The group cover and its local gluing are understood. What is not yet present is an intrinsic frame object over which that cover acts as a lift. The next construction must therefore begin below the group description: from the metric-oriented three-dimensional carrier itself, reconstruct the oriented orthonormal frames basis-free, and only afterwards derive their free and transitive visible-group action. This prevents a convenient group model or the four Task-23 charts from being mistaken for constituents of the new object.

The Conservation of Difficulty has therefore moved from identification, to local gluing, and now to dependency-controlled frame reconstruction. The next question is no longer whether the reconstructed mathematics resembles spin. It is first whether the reconstruction itself can generate, without a privileged reference frame, the frame data that would make a lift question meaningful. Only after that object exists should a subsequent dependency audit ask what additional ingredients would be required to pass from one frame torsor to a frame bundle over a nontrivial base.

A Task-23 module inventory

Module	Lines	Role
SafeBase	352	frozen projection, sign carrier and sign action
Cover	168	four intrinsic domains, openness, cover and minimality
Sections	206	positive local representatives and continuity
Transitions	261	transition signs, regularity, laws, disconnected-overlap counterexample
LocalProduct	130	local product equivalences and homeomorphisms
Reconstruction	228	gluing relation, quotient, homeomorphic recovery of $\mathcal{L}$
GlobalSection	196	local trivialization criterion for a global continuous section
OldDomains	171	exact comparison with old $\text{Dom}(v)$ domains
OldBridges	251	core/extended Bridge separation and integer-label audit
Hierarchy	210	minimal sufficient reconstruction data and negative controls
Comparison	226	comparison-only covering-map classification
Task23	193	55 endpoint aliases and axiom audit

Module	Lines	Role
Total	2592	

B Task-23 decision ledger

Question	Final verdict
Explicit intrinsic open cover of G_{vis}	PROVED
Four-domain coordinate cover minimal within its family	PROVED
Continuous local sections on every cover element	PROVED
Relative local factors exactly in $\{\pm 1\}$	PROVED
Transition factors continuous and locally constant	PROVED
All overlaps connected/preconnected	REFUTED
One sign per pair of domains	REFUTED
Identity/inverse/triple-overlap laws	PROVED
Local product decomposition over each V_i	PROVED AS HOMEOMORPHISM
Global $\mathcal{L}$ reconstructed from local pieces	PROVED AS HOMEOMORPHISM OVER BASE
Old $\text{Dom}(v)$ exactly equal to new local domains	REFUTED / PARTIAL CORRESPONDENCE
Old core Bridge reduces to transition sign	PROVED
Extended Bridge reduces to sign	REFUTED
Parameter-shift integer reduces by parity	PROVED
Central-label full-turn integer reduces by parity	REFUTED; CONSTANT +1
Sign data sufficient for core $\mathcal{L}$	PROVED
Sign data sufficient for full $\mathcal{L}_Z$	REFUTED
Global continuous section iff transitions continuously trivializable	PROVED
Set-theoretic trivialization exists	PROVED
Conventional two-sheeted covering-map classification	PROVED, COMPARISON ONLY
Frame bundle constructed	NO
Spin structure constructed	NO
Physical spin degree of freedom derived	NO

C Failbuild and failed-theorem provenance

Task 23 preserves both engineering failures and the one mathematically important failed intended simplification.

#	Symptom	Diagnosis	Repair / boundary change
1	unknown <code>IsQuotientMap</code>	namespace location in pinned Mathlib	opened <code>Topology</code>
2	old homeomorphism constructor unavailable	constructor renamed in revision	used the revision-correct continuous/open equivalence constructor

#	Symptom	Diagnosis	Repair / boundary change
3	elaboration timeout for inherited product continuity	ambient eight-coordinate multiplication too large for direct unification	proved coordinatewise continuity helper
4	sigma-constructor elaboration failure in openness proof	anonymous constructor could not infer family	added explicit inclusion helper
5	<code>visOf</code> rejected as computable	real-analytic inherited data	marked definition noncomputable
6	Task-23 linter warnings during iteration	redundant tactic branches and unused arguments	cleaned; final Task-23 warning count zero
M	attempted one sign per overlap pair	explicit points of $V_0 \cap V_1$ carry opposite transition signs	theorem boundary changed to locally constant sign functions on overlaps; counterexample preserved

D Open mathematical obligations after Task XXIII

Task 23 closes its requested local reconstruction problem but intentionally leaves several narrower questions unresolved:

1. the connected components of every overlap are not enumerated, and no finite componentwise sign table is produced;
2. it is unknown whether another equally intrinsic cover derived from the axis/parameter reconstruction can be chosen with connected overlaps;
3. an explicit minimal local data set sufficient to reconstruct the extended carrier $\mathcal{L}_Z$ is not constructed;
4. all Task-23 local structure is topological; no smooth or Lie-theoretic local-trivialization statement is proved;
5. uniqueness or naturality of the reconstruction homeomorphism $\mathcal{L}_{\text{loc}} \cong \mathcal{L}$ is not proved;
6. no oriented-frame torsor or frame-bundle object has yet been reconstructed.

The first five are secondary branches. Task 24 should focus only on the sixth.

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